

AI Driven Privacy Preserving Digital Ecosystem for Modernizing Sri Lanka's Paddy Supply Chain

R26 SE 007

Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Sri Lanka's paddy supply chain relies on a regional warehouse network managed by the Paddy Marketing Board (PMB), where critical coordination, including stock monitoring, inter warehouse redistribution, and disaster response, is currently conducted through informal channels such as WhatsApp messages and phone calls. These methods provide no verifiable audit trail, no cryptographic accountability, and no trusted mechanism for validating warehouse reported data. In disaster scenarios such as floods, cyclones, or elephant attacks, Regional Managers must make rapid redistribution decisions without any reliable information infrastructure, leading to stock losses, manipulation risks, and governance failures.

This research identifies a critical knowledge gap: the absence of a privacy preserving, disaster aware, and AI enhanced warehouse coordination system tailored for Sri Lanka's paddy sector. Globally, blockchain and Zero Knowledge Proof (ZKP) technologies have been explored in general supply chains, but no study combines these with Graph Neural Network based fraud detection in an agricultural warehouse context, particularly for disaster coordination in a developing nation.

The proposed solution is a Secure, Disaster Aware Paddy Warehouse Coordination System that integrates: (1) a blockchain based immutable ledger for all warehouse events, (2) role based authentication for warehouses and managers using blockchain native access control, (3) Zero Knowledge Proofs (ZKP) for privacy preserving capacity verification, and (4) Graph Neural Network based anomaly and fraud detection. The system replaces informal coordination with a verifiable, trusted decision making layer without disrupting existing logistics processes.

Technologies employed include Hyperledger Fabric for permissioned blockchain, Circom and SnarkJS for zk SNARK zero knowledge proofs, and PyTorch Geometric for GNN based anomaly detection. Expected outcomes include a provably secure warehouse coordination system, real time reliability scoring for warehouses, and disaster resilient redistribution workflows validated through simulation.

The system delivers measurable social impact by protecting farmer livelihoods, reducing post disaster rice shortages, and strengthening institutional accountability. Commercially, it offers a deployable SaaS platform for national food security agencies. The research directly advances SDG 2 (Zero Hunger), SDG 9 (Infrastructure and Innovation), and SDG 16 (Strong Institutions).

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2. The system shall enforce role based access control, ensuring warehouse supervisors, Regional Managers, and administrators have distinct, non overlapping permission tiers for reading, writing, and endorsing transactions.	
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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
GNN	Graph Neural Network
PMB	Paddy Marketing Board
HARTI	Hector Kobbekaduwa Agrarian Research and Training Institute
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
ZKP	Zero Knowledge Proof
zk SNARK	Zero Knowledge Succinct Non Interactive Argument of Knowledge
Dapp	Decentralized Application
API	Application Programming Interface
SHA	Secure Hash Algorithm
W3C	World Wide Web Consortium
RM	Regional Manager
SaaS	Software as a Service

INTRODUCTION

4.1 Background and Context

Sri Lanka's paddy cultivation system is historically regarded as the cornerstone of national food security, with the country producing approximately 2.7 million metric tons of paddy annually across two main cultivation seasons [1]. The Paddy Marketing Board (PMB), a state institution, operates one of the most extensive regional warehouse networks in the country, tasked with purchasing paddy directly from farmers, storing it in distributed warehouse facilities across districts such as Ampara, Badulla, Kandy, and Polonnaruwa, and redistributing stock to millers and traders in response to supply and demand conditions.

Each Regional Manager (RM) within the PMB is responsible for overseeing a cluster of warehouses within their district. Their responsibilities encompass monitoring warehouse stock levels in metric tons, recording inbound paddy from farmers, managing inter warehouse redistribution, coordinating outbound distribution, and critically, responding to emergency situations caused by natural disasters such as floods, cyclones, and wind damage, as well as wildlife threats including elephant attacks, a recurring challenge in Sri Lanka's paddy producing regions [2].

Despite the critical nature of these operations, the existing information infrastructure supporting warehouse coordination is severely underdeveloped. Regional Managers currently rely on informal communication channels, including telephone calls, WhatsApp messages, and handwritten reports, to exchange stock information, coordinate redistribution, and manage disaster events. While these channels enable rapid communication, they fundamentally lack the properties required for trusted, high stakes logistics: verifiability, auditability, cryptographic accountability, and privacy preserving data sharing [3].

In disaster scenarios, the consequences of this information deficit are severe. When a warehouse suffers damage from flooding or an elephant attack, the Regional Manager must rapidly identify which nearby warehouses can absorb the displaced stock. This decision requires knowing the current available capacity of each candidate warehouse, the reliability of the reported data, and the geographic proximity of alternatives. Currently, there exists no mechanism to cryptographically verify that a warehouse's reported capacity figure is accurate, nor is there an audit trail that records these decisions for post disaster accountability [4].

This research is aligned with the SST (Software Systems and Technologies) Research Cluster at SLIIT, under the Software Engineering specialization, and intersects with High Performance Computing, Algorithms Design, and Distributed Architectures. It contributes to the broader group research project 'AI Driven Privacy Preserving Digital Ecosystem for Modernizing Sri Lanka's Paddy Supply Chain' (Project ID: R26 SE 007), alongside three complementary components addressing digital twin farming, price forecasting, and farmer miller marketplace systems.

This component directly advances UN Sustainable Development Goal 2 (Zero Hunger) by improving food storage resilience and reducing post harvest losses; SDG 9 (Industry, Innovation and Infrastructure) by introducing digital infrastructure for agricultural institutions; and SDG 16 (Peace, Justice and Strong Institutions) by enhancing transparency and accountability in public food system governance.

1.2 Literature Review

1.2.1 Blockchain in Agricultural Supply Chains

Blockchain technology has attracted substantial research interest as a mechanism for introducing transparency and traceability into agricultural supply chains. Antonucci et al. (2019) conducted a systematic literature review of blockchain applications in agri food systems, identifying traceability and provenance verification as the dominant use cases [5]. Their analysis found that while blockchain provides strong immutability guarantees, most implementations focus on consumer facing traceability rather than inter institutional coordination.

Tripoli and Schmidhuber (2018) examined blockchain's potential in agriculture through the FAO lens, concluding that permissioned blockchain architectures (such as Hyperledger Fabric) are better suited to institutional deployments due to their governance controls, compared to public blockchains like Ethereum [6]. This insight directly informs the technology selection for the proposed system, as the PMB operates within a well defined institutional hierarchy.

Ge et al. (2017) and Tian (2016) demonstrated blockchain based traceability for food supply chain tracking, where physical measurements are anchored to blockchain records to create a trusted, tamper evident audit trail [7][8]. However, these systems focused on product quality parameters during transit rather than warehouse capacity management or disaster coordination workflows.

A particularly relevant study by Feng et al. (2023) specifically addresses grain and food supply chain blockchain with Zero Knowledge Proofs [9]. The research demonstrates that ZKP can protect sensitive warehouse capacity information while enabling verifiable claims about stock levels. However, the study was conducted in the context of Chinese grain reserves and does not address disaster time coordination workflows or GNN based anomaly detection.

1.2.2 Zero Knowledge Proofs in Supply Chain Verification

Zero Knowledge Proofs (ZKPs) are cryptographic protocols that allow one party (the prover) to convince another party (the verifier) that a statement is true, without revealing any information beyond the truth of that statement. The foundational work by Goldwasser, Micali, and Rackoff (1989) established the theoretical basis for ZKPs, while subsequent developments, particularly zk SNARKs (Zero Knowledge Succinct Non Interactive Arguments of Knowledge), have made them computationally practical for real world applications [10].

Hassan et al. (2024) conducted a comprehensive review of ZKP applications in blockchain enabled supply chain management, identifying privacy preserving capacity and stock verification as an emerging but underexplored use case [11]. Their analysis found that while ZKPs have been successfully deployed in financial systems (e.g., Zcash) and identity verification, agricultural applications remain nascent. Critically, no existing work applies ZKPs specifically to warehouse capacity claims in agricultural coordination systems.

Kang et al. (2022) applied ZKPs to privacy preserving data sharing in vehicular networks, demonstrating that range proofs (proving a value lies within a specified interval without revealing the exact value) can be generated and verified with acceptable computational overhead for real time

applications [12]. This approach is directly applicable to the warehouse capacity verification challenge, where a warehouse needs to prove ‘available capacity $\geq X$ tons’ without revealing its exact inventory.

1.2.3 Blockchain Based Access Control in Agricultural Systems

Secure authentication and access control within permissioned blockchain networks has been addressed in prior research through built in membership service mechanisms. Hyperledger Fabric’s Membership Service Provider (MSP) provides certificate based identity management natively within the blockchain infrastructure, enabling role based access control without requiring a separate identity infrastructure [13]. Each participant is issued an enrollment certificate by a Fabric Certificate Authority, which is verified for every transaction.

Kang et al. (2022) demonstrated the effectiveness of consortium blockchain membership controls in peer to peer coordination systems, where fine grained access policies enforced at the chaincode level determined which participants could read, write, or endorse specific transaction types [12]. This approach is directly applicable to the proposed system, where warehouse supervisors, Regional Managers, and PMB headquarters require distinct permission tiers without the overhead of a standalone identity infrastructure.

In the agricultural context, Ge et al. (2017) and Tian (2016) used Hyperledger Fabric’s native certificate authority for stakeholder authentication in food supply chain systems, demonstrating that Fabric’s built in MSP is sufficient for multi party institutional coordination without additional identity frameworks [7][8]. The proposed system adopts this same approach, using Fabric MSP enrollment certificates for authentication and transaction signing.

1.2.4 Blockchain Data Integrity and Off Chain Storage

Ensuring data integrity in blockchain based systems without storing large datasets on chain is a well studied challenge. Content addressed off chain storage solutions, where documents are stored externally and their cryptographic hashes anchored on the blockchain, provide a practical approach to maintaining verifiability while controlling storage costs [16]. This technique is fundamental to agricultural supply chain systems where detailed reports and records must be preserved but cannot be stored entirely on chain.

Tian (2016) and subsequent works demonstrated that anchoring SHA 256 hashes of warehouse records on the blockchain creates a tamper evident audit trail: any modification to the off chain document is immediately detectable by comparing the stored hash against the document’s current hash [8]. This mechanism enables the proposed system to maintain full verifiability of warehouse stock reports without the storage costs of recording full documents on chain.

Ge et al. (2017) further showed that combining hash anchored document storage with smart contract enforced access control creates a robust data governance layer suitable for multi party institutional environments [7]. The proposed system adopts this pattern for all warehouse stock reports, redistribution orders, and disaster event records, ensuring that every critical document is both verifiable and access controlled without requiring specialized hardware at warehouse locations.

1.2.5 Graph Neural Networks for Fraud and Anomaly Detection

Graph Neural Networks (GNNs) represent a class of deep learning models specifically designed for graph structured data, where nodes represent entities and edges represent relationships. Since Zhou et

al. (2020) published their comprehensive review of GNN methods [18], GNNs have been successfully applied to fraud detection in financial networks, social network analysis, and traffic prediction, contexts where relational patterns between entities are as informative as individual entity attributes.

Dou et al. (2020) applied GNNs to opinion fraud detection in social networks, demonstrating that fraudulent behavior is characterized by anomalous relational patterns (e.g., unusual connectivity, synchronized activity) that are only detectable through graph level analysis [19]. This principle directly applies to the warehouse coordination context, where collusive misreporting between warehouses, for example, multiple warehouses simultaneously over claiming capacity, may be undetectable through individual analysis but highly visible in graph based pattern analysis.

Liu et al. (2021) applied GNNs to financial transaction fraud detection with state of the art results, establishing that graph based anomaly detection consistently outperforms traditional ML approaches when relational data is available [20]. No existing research applies GNN based anomaly detection to agricultural warehouse networks, representing a clear and significant research gap.

1.2.6 Sri Lanka Specific Agricultural Technology Research

Existing Sri Lanka specific agricultural research has primarily focused on prediction and production optimization. Mendis et al. (2018) from SLIIT proposed ML based harvest and rice demand prediction using RNN/LSTM models, which is the closest precedent work in the Sri Lankan context [21]. However, their work targets production planning rather than warehouse coordination or post harvest logistics.

A systematic review of Sri Lankan agricultural technology literature confirms a complete absence of research combining blockchain, ZKP, or GNN for paddy warehouse management. The PMB's operational challenges, particularly informal disaster coordination, have not been previously addressed through formal technology research, representing both a critical gap and an opportunity for high impact applied research.

1.3 Research Gap

The literature review reveals four interconnected research gaps that collectively define the novelty space of the proposed research:

Gap 1: No Privacy Preserving Warehouse Capacity Verification: Existing agricultural blockchain systems record stock levels as plain data without privacy protection. When warehouses report available capacity, there is no cryptographic mechanism to verify these claims without exposing exact inventory figures. ZKP based range proofs have never been applied to agricultural warehouse capacity verification, leaving a critical gap in trustworthy information sharing between competing or sensitive institutional entities [9][11].

Gap 2: No Disaster Aware Coordination Mechanism for Agricultural Warehouses: Existing blockchain supply chain systems are designed for routine operations; none addresses the specific challenge of rapid, trusted decision making under emergency conditions. The combination of verifiable capacity claims, reliability scoring, and signed redistribution orders in a disaster aware workflow has no precedent in agricultural research.

Gap 3: Absence of GNN Based Anomaly Detection in Agricultural Warehouse Networks: Graph Neural Networks have demonstrated superior performance in fraud detection for financial and social networks [19][20], but have never been applied to agricultural warehouse networks. The relational structure of PMB's warehouse network, with stock flows between warehouses, Regional Manager oversight relationships, and transporter connections, is ideally suited to GNN based anomaly detection.

Gap 4: No Sri Lanka Specific Integrated Solution: No research addresses the specific operational context of Sri Lanka's paddy warehouse system, including its regional distribution, disaster patterns (floods, elephant attacks), informal coordination culture, and institutional structure. A context specific solution that integrates blockchain, ZKP, and GNN for this unique operational environment, specifically designed around disaster coordination workflows and PMB institutional structure, has not been previously proposed.

OBJECTIVES

2.1 Main Objective

To design, develop, and validate a secure, disaster aware, and privacy preserving paddy warehouse coordination system that integrates blockchain based immutable ledgers, Zero Knowledge Proofs (ZKP) for privacy preserving capacity verification, and Graph Neural Network based fraud detection, enabling trusted, rapid, and auditable redistribution decisions during emergency events in Sri Lanka's regional paddy warehouse network.

2.2 Specific Objectives

1. To implement a permissioned blockchain ledger (Hyperledger Fabric) that immutably records all warehouse events, including stock inflows, redistribution, outflows, and disaster damage reports, with cryptographic timestamps, digital signatures, and off chain data hashes.
2. To design and implement a Zero Knowledge Proof (ZKP) mechanism using zk SNARKs that allows warehouses to produce verifiable proofs of available storage capacity (e.g., 'capacity $\geq$ X tons') and reported loss thresholds without revealing exact inventory values.
3. To design and train a Graph Neural Network (GNN) model that represents the warehouse network as a graph and detects anomalous stock patterns, unusual transaction frequencies, and potential collusive misreporting between warehouses, assigning dynamic reliability scores to each warehouse.
4. To design, simulate, and evaluate a complete disaster response workflow, encompassing event recording, ZKP capacity proof submission, GNN informed warehouse ranking, and digitally signed redistribution order issuance, demonstrating measurable improvements in decision trustworthiness and response efficiency over the current informal system.

METHODOLOGY

3.1 Research Approach

This research adopts a Design Science Research (DSR) methodology, which is appropriate for research that creates novel technological artefacts to address real world problems in organizations [22]. DSR prescribes iterative cycles of problem identification, artefact design, development, demonstration, and evaluation. This approach is well suited to the proposed research, as the primary contribution is a functional system that solves a concrete operational problem in the PMB's warehouse network.

The research follows five phases: (1) Problem Formalization and Requirements Elicitation, (2) System Architecture Design, (3) Component Level Development and Integration, (4) Simulation Based Evaluation, and (5) Validation and Reporting. Each phase produces formal artefacts (design documents, models, code, evaluation reports) that are reviewed against criteria derived from the research objectives.

3.2 Phase 1: Problem Formalization and Requirements Elicitation

Domain knowledge will be gathered through structured interviews with PMB Regional Managers and warehouse supervisors, guided by the initial insights from the external supervisor, Mr. Suranga Senanayake. Interview questions will target: current disaster coordination workflows, frequency and types of disaster events, data reporting practices, privacy concerns around sharing exact stock figures, and decision making timelines.

Secondary data will be collected from PMB internal reports and HARTI databases covering historical warehouse stock records, redistribution patterns, and documented disaster incidents. These will inform the design of realistic simulation scenarios for system evaluation.

3.3 Phase 2: System Architecture Design

The overall system architecture follows a layered design comprising five integrated sub systems. Each sub system corresponds directly to one of the specific objectives and is designed for modular implementation and independent testability.

The architecture is built on three principles: (1) privacy by design, ensuring sensitive operational data is never unnecessarily exposed; (2) verifiability, ensuring all critical data points can be cryptographically proven without trust in any single entity; and (3) resilience, ensuring the system remains functional and trustworthy even in disaster conditions with degraded connectivity.

3.4 Phase 3: Component Development

3.4.1 Blockchain Ledger (Hyperledger Fabric)

A permissioned blockchain will be implemented using Hyperledger Fabric, chosen for its support for private data collections, pluggable membership services, and enterprise grade performance. The consortium network will include nodes representing PMB headquarters, Regional Managers, and warehouse facilities. Smart contracts (chaincode) written in Go will implement warehouse event recording logic, ensuring that all inbound/outbound stock movements, redistribution events, and disaster damage reports are immutably recorded with cryptographic timestamps and off chain data hashes (SHA 256).

Off chain data, including detailed stock reports and redistribution decision records, will be stored in a secure document store with their SHA 256 content hashes anchored on chain. This design balances storage efficiency with verifiability: any party can confirm that a stored document has not been altered by comparing its hash against the on chain record.

3.4.2 Zero Knowledge Proof System

The ZKP system will implement zk SNARKs using the Circom circuit language and SnarkJS library, which provide practical performance for range proof generation on commodity hardware. Two primary ZKP circuits will be designed: (1) a Capacity Range Proof, which allows a warehouse to prove that its available capacity is greater than or equal to a specified threshold X , without revealing the exact available capacity or current stock level; and (2) a Loss Threshold Proof, which allows a warehouse to prove that its disaster reported loss is within an acceptable percentage of its reported stock, without revealing exact loss or stock figures.

Proof generation will occur locally at the warehouse node using a lightweight client application. Verification will be performed by the blockchain smart contract, ensuring that only warehouses that have submitted valid proofs can participate in disaster redistribution workflows. The Groth16 proving system will be used for its efficient constant size proofs and fast verification times.

3.4.3 Graph Neural Network Anomaly Detection

The warehouse network will be modelled as a heterogeneous graph where nodes represent warehouses, Regional Managers, and transporters, and edges represent stock movement relationships (with attributes including volume, frequency, and timing), reporting authority relationships, and geographic proximity. Node features will include historical stock patterns, disaster claim frequency, and consistency scores between reported data and prior blockchain records.

A Graph Attention Network (GAT) architecture will be implemented using PyTorch Geometric, chosen for its ability to learn different levels of attention for different neighbor nodes, critical for detecting subtle anomalies in sparse warehouse networks. The model will be trained on historical PMB data supplemented with synthetically generated anomaly scenarios (stock manipulation, capacity over claiming, collusive reporting). The output will be a per warehouse anomaly score and a dynamic reliability rating (0–1 scale) updated after each new stock event.

3.5 Phase 4: Disaster Response Workflow Integration

The disaster response workflow integrates the three core sub systems into a coherent operational process. When a disaster event is reported by a Regional Manager:

1. The disaster event is immutably recorded on the blockchain with the RM's enrolled certificate signature, providing a cryptographically verified audit trail.
2. The affected warehouse's stock is flagged as 'at risk' in the system.
3. Nearby warehouses are requested to submit ZKP capacity proofs for the required volume.
4. The system retrieves each candidate warehouse's current GNN reliability score.
5. Warehouses are ranked by a composite score: $f(\text{distance, verified capacity, reliability score})$.
6. The RM issues a digitally signed redistribution order to the top ranked warehouse.
7. The redistribution event is recorded on the blockchain, completing the audit trail.

3.6 Phase 5: Evaluation Strategy

The system will be evaluated across four dimensions, each with defined performance metrics:

Evaluation Dimension	Metrics	Target
ZKP Performance	Proof generation time, verification time, proof size	< 2s generation, < 100ms verification
GNN Anomaly Detection	Precision, Recall, F1 score on test anomaly scenarios	F1 > 0.85
Blockchain Performance	Transaction throughput (TPS), latency, storage efficiency	> 100 TPS, < 2s finality
Disaster Response Efficiency	Decision time, decision accuracy vs. ground truth, audit trail completeness	50% time reduction vs. informal process

Simulation based evaluation will use synthetic disaster scenarios derived from historical PMB records and expert interviews. For GNN training and evaluation, 80% of data will be used for training and 20% for held out testing, with 5 fold cross validation.

3.7 Tools, Technologies, and Justification

Technology	Tool/Platform	Justification
Blockchain	Hyperledger Fabric	Permissioned, enterprise grade, supports private data collections; superior to Ethereum for institutional deployments
Zero Knowledge Proofs	Circom + SnarkJS (Groth16)	Industry standard zk SNARK implementation; constant size proofs; efficient on chain verification
Off chain Storage	PostgreSQL / AWS S3	Reliable document storage for detailed stock reports; content hashes anchored on chain for verifiability
GNN Framework	PyTorch Geometric	Leading framework for GNN research; extensive support for heterogeneous graphs and GAT architectures
Backend API	Node.js / Python FastAPI	Node.js for Fabric SDK integration; FastAPI for ML model serving with high performance
Frontend	React Native	Cross-platform mobile UI for iOS and Android; NativeWind provides Tailwind-style utility classes for consistent styling

3.8 Data Collection and Ethical Considerations

The research primarily utilizes secondary and institutional data, minimizing ethical risk. Historical warehouse stock records and redistribution data will be obtained from PMB with formal authorization and used exclusively for model training and system validation. All institutional data will be anonymized where necessary and handled in compliance with SLIIT's academic data governance standards. No personal data from individual farmers or employees will be collected.

The research does not involve human experimentation. Expert interviews with PMB personnel will be conducted with informed consent and will not record personally identifiable information. If required by institutional guidelines, formal ethical clearance will be sought prior to data collection.

HIGH LEVEL SYSTEM ARCHITECTURE

4.1 Overall System Overview

The proposed system is designed as a four layer architecture that integrates blockchain infrastructure, cryptographic verification, and AI based analytics into a unified operational platform for PMB warehouse coordination. The system is positioned as a trusted middleware layer that enhances existing logistics processes without replacing them.

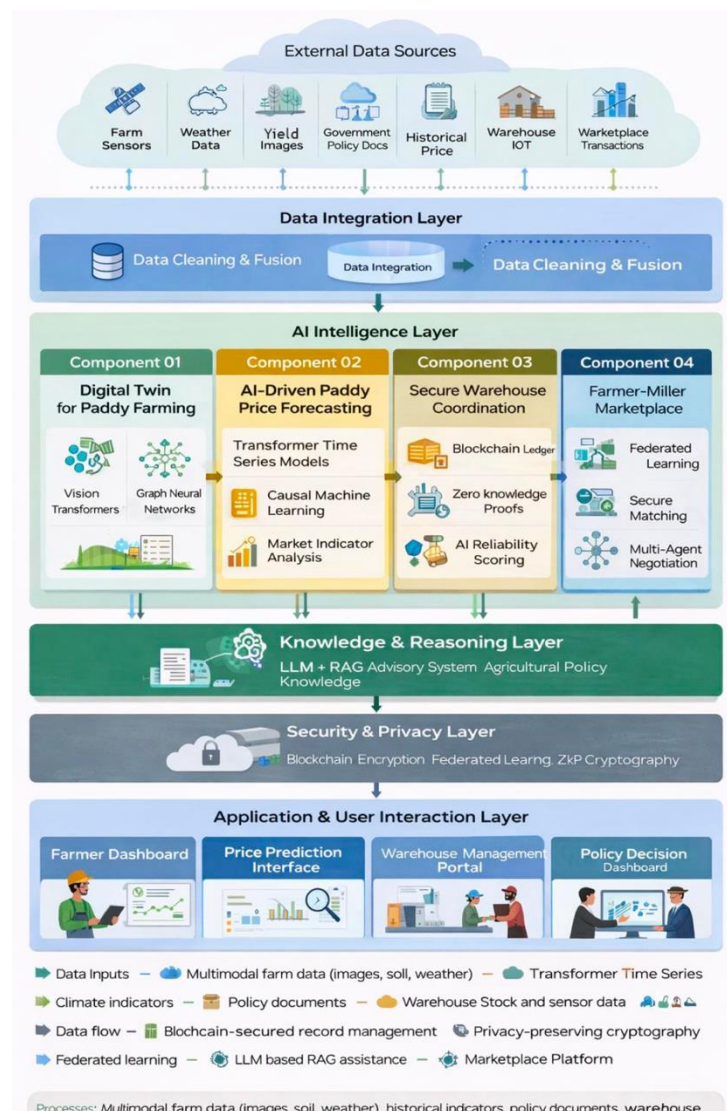


Figure 1 System Diagram of Components – AI Driven Privacy Preserving Digital Ecosystem for Modernizing Sri Lanka's Paddy Supply Chain

4.2 Component Architecture

4.2.1 Data Layer (Blockchain Ledger)

The data layer is centred on the Hyperledger Fabric blockchain network, which maintains the immutable ledger of all warehouse events. Smart contracts enforce data validation rules and role based access control policies. Detailed stock reports and redistribution records are stored off chain in a secure document store, with their SHA 256 content hashes anchored on chain, enabling verifiability without excessive storage costs.

4.2.2 Verification Layer (ZKP System)

The verification layer implements the zero knowledge proof protocols for capacity and loss verification. When a warehouse needs to participate in disaster redistribution, it generates a ZKP proving its available capacity meets the required threshold. The proof is submitted to the blockchain, where the smart contract verifies it using the corresponding verification key. Only warehouses with valid, current capacity proofs can be ranked for redistribution, ensuring that redistribution decisions are based on cryptographically verified capacity claims.

4.2.3 Analytics Layer (GNN Anomaly Detection)

The analytics layer continuously analyses the warehouse network graph to detect anomalous patterns and maintain dynamic reliability scores. The graph is updated in real time as new blockchain events are recorded, with the GNN model re computing anomaly scores after each significant event. The reliability scores are made available to the Regional Manager dashboard and are integrated into the warehouse ranking formula for disaster redistribution decisions.

4.2.4 Coordination Layer (Disaster Response Workflow)

The coordination layer integrates all underlying components into the operational disaster response workflow. A React Native mobile application with NativeWind styling provides Regional Managers with real time visibility of warehouse status, active ZKP verified capacity claims, GNN reliability scores, and the ranked list of candidate redistribution warehouses, backed by a Node.js REST API. Signed redistribution orders are generated through the mobile interface and recorded on the blockchain upon RM confirmation.

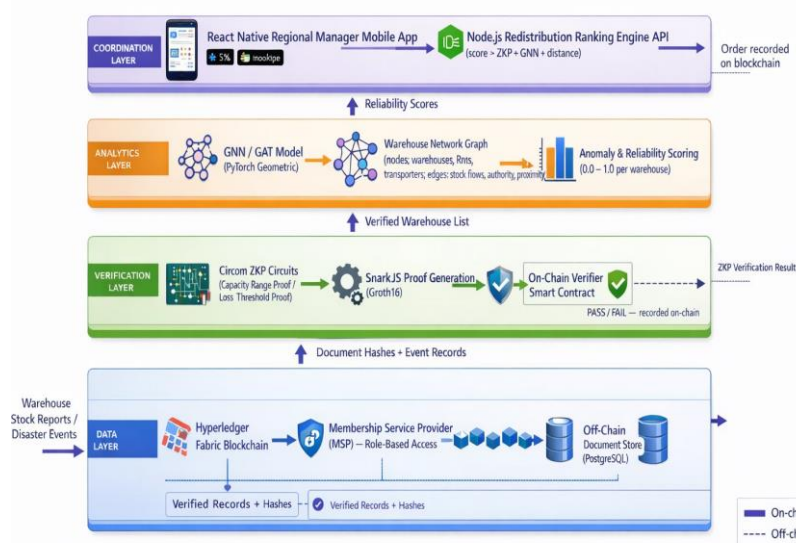


Figure 2 System Diagram for the Secure, Disaster Aware Paddy Warehouse Coordination Component (Component 03)

4.3 Data Flow Description

Under normal operations, warehouses periodically submit stock reports (authenticated via their Fabric enrollment certificate) to the blockchain ledger. The GNN model continuously processes the updated warehouse network graph and refreshes reliability scores. All data is accessible to authorized Regional Managers through the dashboard.

Under disaster conditions, the workflow described in Section 3.5 is activated. The system transitions from passive monitoring to active coordination mode, with ZKP proof requests issued to candidate warehouses, reliability scores prominently weighted in the ranking algorithm, and time stamped redistribution decisions permanently recorded on the blockchain for audit purposes.

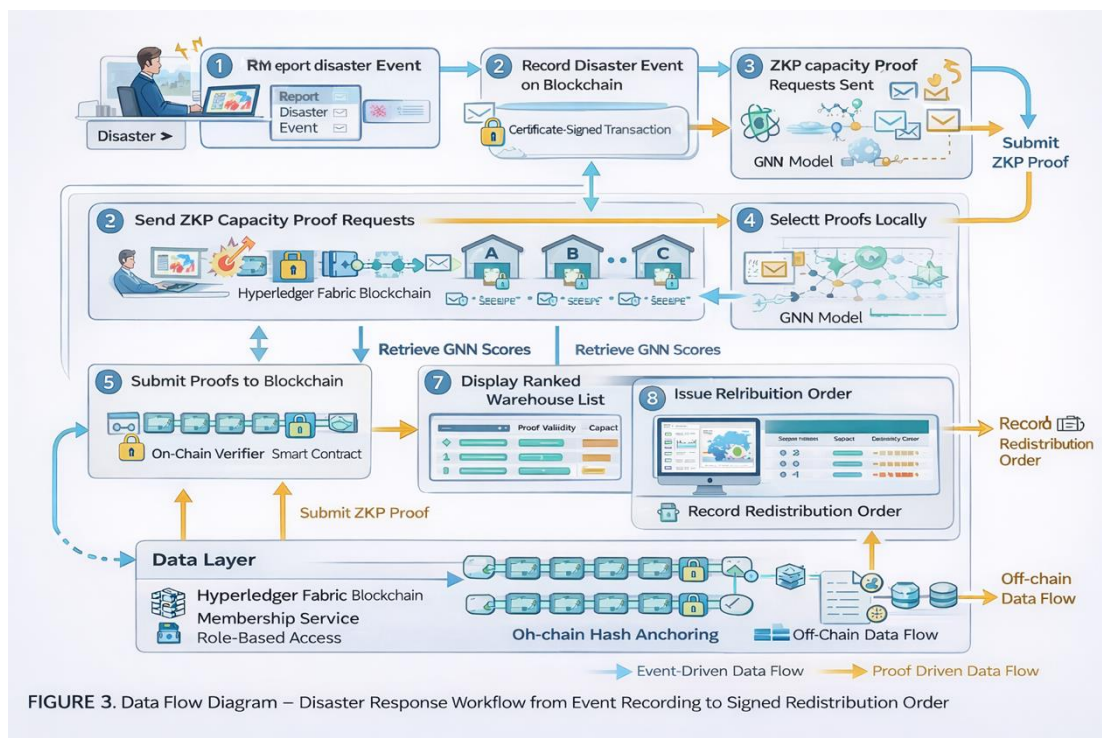


Figure 3 Data Flow Diagram – Disaster Response Workflow from Event Recording to Signed Redistribution Order

4.4 Scalability and Security Considerations

The Hyperledger Fabric network is designed for horizontal scalability, with additional peer nodes deployable as the number of participating warehouses grows. The ZKP proof generation is computationally intensive (approximately 2–5 seconds for Groth16 circuits of the required size) but is performed locally at warehouse nodes, avoiding bottlenecks at the central network level. The GNN model is designed for incremental graph updates rather than full retraining, ensuring real time anomaly detection remains feasible as the network scales.

Security considerations include the ZKP trusted setup ceremony (using multi party computation to prevent single point of trust failure), Fabric chaincode security auditing, and encryption of off chain document storage. Fabric private data collections ensure that sensitive stock reports are accessible only to authorized participants, while the blockchain ledger provides a publicly auditable record of all transaction hashes and outcomes.

USER REQUIREMENTS

5.1 Stakeholder Identification

Stakeholder	Role	Primary Interaction
Regional Manager (RM)	Oversees warehouse cluster; makes redistribution decisions	Dashboard, redistribution orders, proof verification
Warehouse Supervisor	Reports stock events; submits ZKP capacity proofs; confirms redistribution instructions	Stock reporting interface, ZKP proof client
PMB Headquarters	System administrator; manages network certificates and access control	Certificate authority management, audit dashboard
Auditors	Post disaster and routine audits	Blockchain audit trail access, report generation
System Administrator	Manages blockchain network nodes and system infrastructure	Node management, system monitoring, certificate management

5.2 Functional Requirements

FR1, Identity Management

FR2, Blockchain Ledger

8. The system shall record all warehouse stock inflows, outflows, inter warehouse redistribution events, and disaster damage reports as immutable blockchain transactions.
9. Each blockchain transaction shall include: timestamp, participant identifier, transaction type, data hash, and digital signature for non repudiable attribution.
10. The system shall store detailed report data in a secure off chain document store and record its SHA 256 content hash on chain for verifiability.
11. The system shall allow authorized entities to query the complete transaction history of any warehouse.

FR3, Zero Knowledge Proof Verification

12. The system shall allow a warehouse to generate a ZKP proving its available capacity meets a specified threshold, without revealing exact inventory values.
13. The system shall allow a warehouse to generate a ZKP proving its disaster loss claim falls within an acceptable threshold.
14. The blockchain smart contract shall verify submitted ZKPs and record verification results immutably.
15. The system shall only admit warehouses with valid ZKP capacity proofs to the disaster redistribution ranking process.

FR4, GNN Anomaly Detection

16. The system shall model the warehouse network as a graph and apply GNN based analysis to detect anomalous stock patterns and potential fraud.
17. The system shall compute and maintain a dynamic reliability score (0.0–1.0) for each warehouse, updated after each stock event.
18. The system shall flag warehouses exhibiting anomalous behavior for manual review by the Regional Manager.

FR5, Disaster Response Workflow

19. The system shall support a structured disaster response workflow: event recording → ZKP proof solicitation → warehouse ranking → RM decision → signed redistribution order.
20. The system shall rank candidate warehouses by a composite score incorporating geographic distance, verified capacity, and GNN reliability score.
21. The system shall generate digitally signed redistribution orders that are immutably recorded on the blockchain.
22. The system shall produce a complete, blockchain anchored audit report for every disaster event.

5.3 Non Functional Requirements

Category	Requirement	Target Specification
Performance	ZKP Proof Generation	< 2 seconds on standard warehouse node hardware
Performance	ZKP Proof Verification	< 100 milliseconds (blockchain smart contract)
Performance	Blockchain Transaction Finality	< 2 seconds for Fabric permissioned network
Performance	Blockchain Throughput	> 100 TPS to support concurrent warehouse submissions
Security	Data Confidentiality	Exact stock values never exposed without authorization; ZKP ensures verifiability without disclosure
Security	Non repudiation	All transactions signed with Fabric enrollment certificates; cryptographically non repudiable
Security	Tamper Evidence	Blockchain immutability ensures no historical record can be altered post submission
Reliability	System Availability	> 99.5% uptime for blockchain and dashboard services
Usability	Dashboard Interface	Task completion rate > 90% for RM users without prior system training
Scalability	Network Scale	Supports up to 200 warehouse nodes without architectural changes

COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market is government food security agencies and agricultural marketing boards in Sri Lanka and comparable developing nations facing similar warehouse coordination and food security challenges. The immediate customer is the Sri Lanka Paddy Marketing Board (PMB), which oversees approximately 60+ regional warehouses and represents a direct institutional partner with validated operational need.

The secondary target market includes regional agricultural cooperatives, rice millers' associations, and private grain storage companies that operate distributed warehouse networks and face analogous information integrity and coordination challenges. Globally, similar opportunities exist in Myanmar, Bangladesh, India (state level grain agencies), and sub Saharan African nations with national grain reserve systems.

6.2 Value Proposition

The system delivers distinct and measurable value across three dimensions. For Regional Managers, it replaces informal, unverifiable coordination with a trusted, auditable decision support system that significantly reduces the risk of poor redistribution decisions and post disaster stock losses. For PMB headquarters and auditors, it provides cryptographically guaranteed audit trails that eliminate disputes about who reported what and when. For farmers and policymakers, it strengthens institutional confidence in the food storage system, reducing market speculation triggered by disaster uncertainty.

The privacy preserving design, particularly the ZKP based capacity verification, addresses a critical barrier to data sharing in institutional contexts: warehouses can prove their capacity meets redistribution requirements without exposing sensitive operational data to competitors or unauthorized parties.

6.3 Revenue Model

The commercialization strategy follows a tiered Software as a Service (SaaS) model for government and institutional clients:

Tier	Features	Pricing Model
Basic	Blockchain ledger, role based access control, RM dashboard	Per warehouse subscription (LKR 15,000/month)
Standard	All Basic + ZKP capacity verification + GNN reliability scoring	Per warehouse subscription (LKR 35,000/month)
Enterprise	All Standard + GNN analytics + custom reporting + API access	Institutional licensing (negotiated annually)

Additional revenue streams include: one time implementation fees, training and onboarding services, custom smart contract development for institutional specific workflows, and data analytics reports for government policy planning.

6.4 Competitive Advantage

The system's primary competitive advantage is its unique combination of privacy preserving cryptography (ZKP) and AI based fraud detection (GNN) within a permissioned blockchain platform

purpose built for agricultural warehouse coordination. No commercially available warehouse management system offers ZKP based capacity verification, and no existing product targets agricultural warehouse coordination in disaster prone developing nations. This combination of technical novelty and domain specificity creates a strong first mover advantage in an underserved market.

The validated institutional partnership with PMB and the external supervisorship of a serving PMB Regional Manager provides an authenticity and credibility advantage that generic supply chain platforms cannot replicate. This relationship accelerates trust building with prospective government clients and provides direct access to real operational requirements for continuous improvement.

6.5 Intellectual Property Considerations

The ZKP circuit designs and GNN model architecture developed in this research may be eligible for software copyright protection and potentially a provisional patent for the novel combination of disaster aware warehouse coordination with privacy preserving cryptographic verification. The research findings will be published in peer reviewed venues, establishing academic priority while demonstrating technical credibility to prospective commercial partners. Commercialization will be pursued through SLIIT's technology transfer framework.

BUDGET AND JUSTIFICATION

The budget is primarily driven by cloud hosting costs for the Hyperledger Fabric network and GNN model training, which require dedicated compute resources. No specialized hardware is required, as the system operates entirely through software on cloud and development machines. All core software tools are open source, eliminating licensing costs.

The technology stack was deliberately selected to maximise academic reproducibility and minimise cost, while maintaining production grade quality:

- Hyperledger Fabric runs locally via Docker Compose on a development machine no paid cloud nodes are required for prototype development and evaluation.
- Circom and SnarkJS (zk SNARK toolchain) are fully open source under MIT/LGPL licences with no usage fees.
- GNN training (PyTorch Geometric) is performed on Google Colab's free T4 GPU tier or Kaggle Notebooks, which provide sufficient compute for the prototype dataset scale.
- PostgreSQL off chain storage is hosted on Supabase's free tier (500 MB), adequate for warehouse event records at prototype scale.
- The React Native mobile application is built with Expo for cross-platform deployment; backend Node.js APIs on Render's free tier, both sufficient for academic evaluation and demonstration.

SLIIT's institutional library access covers IEEE Xplore and ACM Digital Library, eliminating any research paper access costs.

7.1 Detailed Budget Breakdown

Category	Item / Description	Tool / Source	Est. Cost	Justification
Blockchain Infrastructure	Hyperledger Fabric Network	Local Docker / GitHub Codespaces (free tier)	LKR 0	Hyperledger Fabric runs locally via Docker Compose. No cloud hosting required during development.
	Smart Contract (Chaincode) Development	Node.js SDK – open source (Apache 2.0)	LKR 0	Fabric SDK for Node.js is fully open source with no licensing cost.
	Off chain Storage (PostgreSQL)	PostgreSQL – open source / Supabase free tier	LKR 0	Supabase provides 500 MB free PostgreSQL storage, sufficient for prototype scale warehouse event data.
Zero Knowledge Proof System	ZKP Circuit Development (Circom)	Circom – open source (MIT); SnarkJS – open source (LGPL)	LKR 0	Circom and SnarkJS are the industry standard open source toolchain for zk SNARK development.
	ZKP Proof Generation & Verification	SnarkJS (browser/Node.js) – free	LKR 0	Proof generation runs on development machine. Verification is on chain within Hyperledger Fabric chaincode.
AI / GNN Model Development	GNN Model Training (PyTorch Geometric)	Google Colab (free GPU tier) / Kaggle Notebooks	LKR 0	Google Colab T4 GPU (free tier) and Kaggle Notebooks provide sufficient compute for GAT model training on a synthetic warehouse graph dataset.
	PyTorch Geometric Framework	PyTorch Geometric – open source (MIT)	LKR 0	Open source GNN research framework with no licensing cost.
	Synthetic Dataset Generation (OpenAI API)	OpenAI GPT 4o / GPT 4o mini API	~LKR 1,500	Used to generate realistic synthetic paddy warehouse transaction records and anomaly scenarios for GNN training. GPT 4o mini pricing (~\$0.15/1M tokens) keeps this minimal.
Frontend & Backend Development	React Native / NativeWind	React Native – open source (MIT); NativeWind – open source (MIT)	LKR 0	React and all associated UI libraries (Tailwind CSS, shadcn/ui) are open source.

	Backend API (FastAPI / Node.js)	FastAPI (Python) – open source; Node.js – open source	LKR 0	Both frameworks are open source with no licensing cost.
Development Tools & Hosting	Version Control & CI/CD	GitHub (free for students / public repos)	LKR 0	GitHub Student Developer Pack provides free access to GitHub Actions CI/CD pipelines.
	IDE & Development Environment	VS Code – free; Docker Desktop – free	LKR 0	All development tooling is freely available.
	Project Demo Hosting	Vercel (free tier) / Render (free tier)	LKR 0	Frontend dashboard can be deployed on Vercel free tier. Backend APIs on Render free tier (sufficient for demo/evaluation purposes).
	Research & Documentation	Google Scholar, IEEE Xplore (SLIIT library access), Overleaf free tier	LKR 0	SLIIT institutional access covers IEEE Xplore and ACM Digital Library subscriptions.
Contingency	Additional API usage (if synthetic dataset needs expansion)	OpenAI API – pay per use	~LKR 1,500	Reserved buffer for additional LLM assisted data generation if the initial dataset requires augmentation during GNN validation.
TOTAL ESTIMATED PROJECT COST			~LKR 3,000	All costs arise solely from OpenAI API usage for synthetic data generation. All other components use free, open source, or free tier cloud tools.

Legend:

■ Free / LKR 0	■ Minor API cost
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WORK BREAKDOWN STRUCTURE (WBS)

8.1 WBS Overview

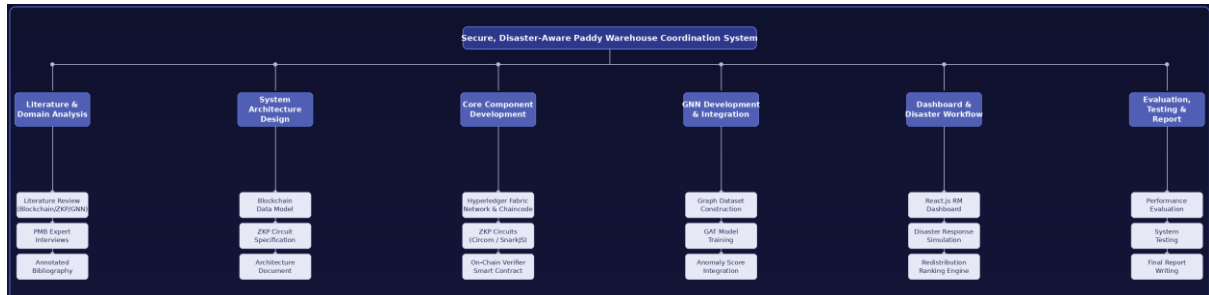


Figure 4 Work Breakdown Structure (WBS) – Secure, Disaster Aware Paddy Warehouse Coordination System

GANTT CHART

Task / Activity	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
P1: Literature Review & Proposal Preparation (Feb 1 – Mar 14)											
Initial Research Topic Exploration											
Literature Review (Blockchain, ZKP, GNN)											
Agricultural Supply Chain Research											
Domain Analysis of PMB Warehouse System											
Problem Definition & Research Gap ID											
Write Research Proposal											
Prepare Proposal Presentation											
Milestones											
♦ Proposal Submission											
♦ Proposal Presentation											
P2: System Design & Planning (Mar 19 – Apr 10)											
Requirement Analysis											
System Architecture Design											
Blockchain Data Model Design											
Smart Contract & ZKP Circuit Design											
P3: Core System Development (Apr 11 – May 30)											
Setup Hyperledger Fabric Network											
Develop Blockchain Smart Contracts											
Implement Warehouse Event Recording											
Off Chain Storage & ZKP Circuits											
ZKP Verifier Integration											
Milestones											
♦ Progress Presentation 1 (4–6 May)											
♦ Checklist Submission (6 May)											
P4: AI / GNN Model Development (Jun 1 – Jul 5)											

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APPENDICES

Appendix A: Risk Analysis and Mitigation Plan

Risk	Likelihood	Impact	Mitigation Strategy
Limited access to real PMB warehouse data	Medium	High	Use synthetic/anonymized datasets; supplement with HARTI public records
ZKP proof generation too slow for real time use	Low	Medium	Pre compute proofs; use Groth16 for constant time verification; cache recent proofs
Hyperledger Fabric network setup complexity	Medium	Medium	Use Docker based Fabric test network; leverage official SLIIT lab resources
Insufficient labeled anomaly data for GNN training	High	High	Generate synthetic anomaly scenarios; use semi supervised GNN training techniques
ZKP proof generation complexity and trusted setup overhead	Low	Medium	Use pre built Circom circuit templates; conduct trusted setup using established multi party computation tools; validate on testnet before mainnet deployment
Scope creep across five integrated sub systems	Medium	High	Strict WBS adherence; modular development with independent testability per component

Appendix B: Ethical Considerations and Data Governance Statement

This research does not involve human participants in experimental procedures. The primary ethical considerations relate to institutional data privacy and AI driven decision support in public food system governance.

Data Privacy: All institutional data obtained from the PMB or HARTI will be used strictly with formal written authorization. Data will be anonymized at the record level before use in model training. No personally identifiable information (PII) relating to farmers, warehouse staff, or institutional employees will be collected or processed.

Informed Consent: Expert interviews with PMB personnel will be conducted with explicit informed consent. Participants will be briefed on the research purpose, their right to withdraw, and how their responses will be used. Interview data will be paraphrased and anonymized in all published outputs.

AI Decision Support: The GNN based reliability scoring system is designed as a decision support tool, not an autonomous decision maker. All redistribution decisions remain under human (Regional Manager) authority. The system provides ranked recommendations; final approval authority rests entirely with authorized PMB personnel.

Ethical Clearance: Formal ethical clearance will be sought from SLIIT's Research Ethics Committee prior to any institutional data acquisition, in accordance with SLIIT's academic research governance framework.